Chemical Substance Synthesis

Problem Description

In a laboratory, chemical substance synthesis is an important area of study.

There are N types of substances (numbered 1 to N). The preparation time for the i-th substance from start to completion is Ti.

Two substances can react to form a new substance. The time required for a reaction is determined by the longer preparation time of the two substances involved.

For example: if substance A takes 3 days to prepare and substance B takes 8 days, then the reaction between A and B takes 8 days.

Each reaction produces a substance that is still among these N types.

Initially, the laboratory possesses M types of substances (in unlimited quantities, allowing multiple reactions simultaneously).

Multiple reactions can be carried out in parallel.

The task is to determine the minimum number of days required to obtain a given target substance.

Example: Suppose there are 4 substances A,B,C,D with preparation times:

A: 5 days

B: 6 days

C: 3 days

D: 8 days

Initially, the lab has substances A and B.

The target substance is D.

Reaction rules:

A+B→C

A+C→D

Optimal process:

Day 1 to Day 6: A reacts with B to produce C (time determined by B)

Day 7 to Day 11: A reacts with C to produce D (time determined by A)

Total time required: 11 days.

It is guaranteed that the target substance can be obtained through reactions.

It is NOT guaranteed that a pair like A and B produces only one substance C (e.g., both A×B→C andA×B→D may exist).

It is NOT guaranteed that a substance D has only one production method that A reacts with (e.g., both A×B→D and A×C→D may exist).

Duplicate reaction rules may appear in the input.

Input and Output

Input Format

The first line contains 4 integers N,M,K,T:

N: total number of substances (numbered 1 to N)

M: number of initially available substance types

K: number of reaction rules

T: target substance index

The second line contains N integers, where the i-th integer is Ti, the preparation time of substance i.

The third line contains M integers representing the initially available substances Kj (all distinct).

The next K lines each contain three integers A,B,C, meaning: substances A and B can react to produce substance C.

Output Format

Output a single integer representing the minimum time required to obtain the target substance.

Constraints

$1 \le N \le 2000$

$2 \le M \le N$

$1 \le K \le 100000$

$1 \le T \le N$

$1 \le T_i \le 100$

$1 \le K_j \le M$

Sample

Sample Input 1:

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6 2 4 6
5 3 4 6 4 9
1 2
1 2 3
1 3 4
2 3 5
4 5 6
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Sample Output 1:

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16
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Explanation:

Day 1 to Day 5: substances 1 and 2 react to produce substance 3.

Day 6 to Day 10: substances 1 and 3 react to produce substance 4.

Day 6 to Day 9: substances 2 and 3 react to produce substance 5.

Day 11 to Day 16: substances 4 and 5 react to produce substance 6.

Total time required: 16 days.